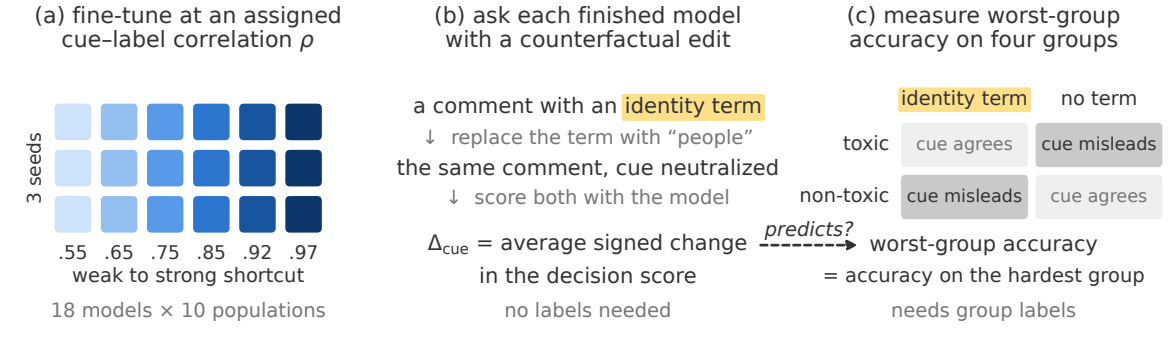
(a) fine-tune at an assigned cue–label correlation ρ
3 seeds
.55 .65 .75 .85 .92 .97
weak to strong shortcut
18 models × 10 populations
(b) ask each finished model with a counterfactual edit
a comment with an identity term
↓ replace the term with "people"
the same comment, cue neutralized
↓ score both with the model
Δcue = average signed change in the decision score
no labels needed
predicts?
(c) measure worst-group accuracy on four groups
identity term
no term
toxic
cue agrees
cue misleads
non-toxic
cue misleads
cue agrees
worst-group accuracy
= accuracy on the hardest group
needs group labels